

DKTE College Training

Agenda

- Linked List
- Binary Tree

Linked List

```
// without header node
class LinkedList {
    Node head = null;

    void addLast(int val) {
        Node nn = new Node(val);
        if(head == null)
            head = nn;
        else {
            Node trav = head;
            while(trav.next != null)
                trav = trav.next;
            trav.next = nn;
        }
    }
    void display() {
        Node trav = head;
        while(trav.next != null) {
            System.out.println(trav.val);
            trav = trav.next;
        }
    }
}
```

```
// with header/dummy node
class LinkedList {
    // head is initialized to empty/dummy node
    Node head = new Node(0);

    void addLast(int val) {
        Node nn = new Node(val);
        Node trav = head;
        while(trav.next != null)
            trav = trav.next;
        trav.next = nn;
    }
    void display() {
        Node trav = head.next;
        while(trav.next != null) {
```

```
        System.out.println(trav.val);
        trav = trav.next;
    }
}
```

Assignments

1. <https://leetcode.com/problems/binary-tree-preorder-traversal>
2. <https://leetcode.com/problems/binary-tree-postorder-traversal>
3. <https://leetcode.com/problems/binary-tree-inorder-traversal>
4. <https://leetcode.com/problems/insert-into-a-binary-search-tree>
5. <https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree>
6. <https://leetcode.com/problems/implement-stack-using-queue>
7. <https://leetcode.com/problems/implement-queue-using-stacks>
8. <https://leetcode.com/problems/find-consecutive-integers-from-a-data-stream>